

Physics-Guided Loss Functions for Low-Light Image Restoration

1. Introduction and Motivation

Low-light image restoration is a fundamental problem in computational photography, particularly relevant for mobile and handheld devices where dim lighting conditions are common. Current state-of-the-art deep learning methods, including DarkIR, have demonstrated impressive results by learning end-to-end mappings from degraded to clean images. However, these approaches often treat the restoration process as a black-box optimization problem, lacking explicit modeling of the underlying physical processes that cause image degradation.

This research proposes to bridge the gap between data-driven deep learning and physics-based image formation models by introducing a suite of differentiable loss functions that encode known physical priors. Specifically, we focus on three key physical phenomena in low-light photography:

1. **Retinex decomposition:** The separation of illumination and reflectance components, where illumination varies smoothly while reflectance contains scene details
2. **Blur formation:** The convolution of sharp images with point spread functions (PSF), particularly from camera shake during long exposures
3. **Frequency domain properties:** The relationship between illumination (low-frequency amplitude) and detail (high-frequency phase) in the Fourier domain

2. Problem Statement

Current deep learning methods for low-light restoration suffer from several key limitations:

2.1 Lack of Physical Grounding

Existing methods optimize networks using generic losses (L1, L2, perceptual) that do not explicitly model the physics of image formation. This results in:

- Networks that may learn spurious correlations rather than true underlying physical relationships

- Poor generalization to degradation types not well-represented in training data
- Difficulty in understanding what the network has learned and why it fails in certain cases

2.2 Suboptimal Illumination Modeling

While methods like RetinexNet and DarkIR implicitly separate illumination and reflectance through architecture design, they lack explicit regularization to enforce physically meaningful decompositions. The Retinex theory states that:

- Illumination should be spatially smooth (slowly varying)
- Reflectance should contain sharp edges and textures
- The observed image is the product $I = R \odot L$ (element-wise multiplication)

Current methods do not enforce these constraints during training, leading to unrealistic decompositions and artifacts.

2.3 Blur Handling Without PSF Awareness

Deblurring networks learn to sharpen images through data-driven feature extraction, but they lack explicit modeling of the blur formation process:

$$\text{Blurred} = \text{Sharp} \otimes \text{PSF}$$

where $\otimes$ denotes convolution and PSF is the point spread function. Without this physical model:

- Networks may introduce unnatural sharpening artifacts
- Gradient statistics of enhanced images may not match natural image gradients
- Performance degrades on blur types not seen during training

2.4 Limited Frequency Domain Exploitation

While DarkIR and other methods use Fourier transforms in the architecture, they primarily focus on amplitude enhancement and neglect phase information. However:

- Phase contains critical edge and detail information
- Low-light images have degraded amplitude (brightness) but relatively preserved phase
- Joint amplitude-phase consistency should be enforced to prevent detail loss during enhancement

3. Research Objectives

The primary objective is to develop a suite of physics-guided loss functions that can be integrated into existing low-light restoration architectures (such as DarkIR) to improve image quality, generalization, and physical plausibility. Specific objectives include:

3.1 Retinex Decomposition Loss

- Design a differentiable loss function that enforces Retinex-compliant illumination-reflectance decomposition
- Incorporate smoothness regularization on illumination components
- Ensure reconstruction consistency: $I = R \odot L$
- Enable the network to learn physically meaningful illumination maps without requiring explicit decomposition networks

3.2 Blur-Aware Gradient Loss

- Develop a gradient consistency loss that accounts for blur formation through implicit PSF estimation
- Ensure that sharpened outputs, when re-blurred with estimated kernels, match the gradient statistics of the input
- Investigate lightweight PSF estimation methods (Gaussian approximation, FFT-based deconvolution)
- Complement existing edge losses by focusing on blur-formation physics rather than generic edge sharpness

3.3 Phase-Enhanced Frequency Loss

- Extend DarkIR's amplitude-only frequency loss to include phase gradient consistency
- Formulate phase loss to avoid wrapping issues (using cosine/sine representations or directional gradients)
- Balance amplitude enhancement with phase preservation to maintain detail integrity
- Investigate multi-scale frequency losses (different frequency bands weighted differently)

4. Methodology

4.1 Foundation: Building Upon DarkIR

Our physics-guided loss functions are designed to enhance the DarkIR architecture without requiring structural modifications, enabling direct integration into the existing framework. We briefly describe DarkIR's design to contextualize how our losses complement its capabilities.

DarkIR Architecture Overview:

DarkIR follows an asymmetric encoder-decoder structure with task-specific blocks:

- **Encoder (EBlocks):** Focuses on low-light enhancement using frequency domain operations. Each EBlock contains:
 - Spatial Attention Module (SpAM) with gating mechanisms
 - Frequency MLP (FreMLP) operating on Fourier amplitude to enhance illumination
 - Features are progressively downsampled (2×, 4×, 8× resolution)
- **Decoder (DBlocks):** Focuses on deblurring and detail recovery in spatial domain. Each DBlock contains:
 - Dilated Spatial Attention Module (Di-SpAM) with fixed dilations [1, 4, 9] for large receptive fields
 - Gated Feed-Forward Network for spatial transformations
 - Features are progressively upsampled back to full resolution
- **Intermediate Supervision:** The encoder produces a low-resolution enhanced image ($\hat{x} \downarrow 8$) at 8× downsampled resolution, which is supervised by L_{lol} to ensure proper illumination enhancement before deblurring.
- **Current Loss Function:**

$$L_{total} = \lambda_{pixel} \cdot L1 + \lambda_{percep} \cdot L_{LPIS} + \lambda_{edge} \cdot L_{edge} + \lambda_{lol} \cdot L_{lol}$$

where $L1$ ensures pixel fidelity, L_{LPIS} provides perceptual quality, L_{edge} preserves high-frequency details, and L_{lol} guides the encoder's illumination enhancement.

Why DarkIR is an Ideal Testbed:

1. **Explicit frequency domain processing:** FreMLP already operates in Fourier space, making our phase-enhanced loss a natural extension
2. **Asymmetric design:** Separate encoder/decoder responsibilities align with our physics-based decomposition (illumination in encoder, blur in decoder)
3. **Intermediate outputs:** The $\hat{x} \downarrow 8$ provides a natural point to apply Retinex constraints

4. **State-of-the-art baseline:** Strong performance (27.3 dB on LOLBlur) ensures gains from physics losses are meaningful

Our approach **augments rather than replaces** DarkIR's architecture, integrating seamlessly into the training pipeline with minimal computational overhead.

4.2 Physics-Guided Loss Functions

We introduce three differentiable loss components that encode physical priors about illumination, blur, and frequency structure. These losses operate on network predictions without requiring architectural changes.

4.2.1 Retinex Decomposition Loss (L_{retinex})

Motivation: Current methods implicitly learn illumination-reflectance separation but lack explicit regularization to enforce physically meaningful decompositions. Retinex theory states that observed intensity I is the product of smooth illumination L and detailed reflectance R .

Formulation:

Given predicted enhanced image I_{pred} , we extract components:

$$L = \text{MaxPool2D}(I_{\text{pred}}, \text{kernel}=15)$$

$$R = I_{\text{pred}} / (L + \epsilon)$$

The loss enforces two physical constraints:

1. **Illumination Smoothness:** Light varies gradually in space

$$L_{\text{smooth}} = ||\nabla L||_1,$$

where ∇ denotes spatial gradients (computed via Sobel or finite differences)

2. **Reconstruction Consistency:** The decomposition must be reversible

$$L_{\text{consist}} = ||I_{\text{pred}} - (R \odot L)||_1,$$

where $\odot$ is element-wise multiplication

Combined:

$$L_{\text{retinex}} = \lambda_{\text{smooth}} \cdot L_{\text{smooth}} + \lambda_{\text{consist}} \cdot L_{\text{consist}}$$

Why This Helps:

- Prevents unrealistic illumination maps (e.g., high-frequency artifacts in L)
- Ensures reflectance contains scene detail, not lighting
- Acts as regularization against overfitting to illumination-specific patterns
- Can be applied to final output AND intermediate $x_{\downarrow 8}$ for stronger supervision

Implementation Details:

- Max pooling with $k=15$ chosen empirically (typical illumination scale)
- Alternative: Replace with learned lightweight decomposition head if max-pooling proves insufficient
- $\varepsilon = 1e-4$ for numerical stability
- Initial weights: $\lambda_{\text{smooth}} = 0.1, \lambda_{\text{consist}} = 1.0$

4.2.2 Blur-Aware Gradient Loss ($L_{\text{blur_grad}}$)

Motivation: Deblurring networks sharpen images via learned features but lack explicit modeling of the blur formation process: $y = x \otimes k$. Without PSF awareness, networks may produce unnatural sharpening or introduce artifacts.

Formulation:

For network prediction x_{pred} (deblurred) and blurry input y , we enforce gradient consistency under re-blurring:

$$L_{\text{blur_grad}} = ||\nabla(x_{\text{pred}} \otimes k_{\text{est}}) - \nabla y||_2^2$$

where k_{est} is an estimated blur kernel and ∇ computes image gradients.

PSF Estimation Strategy (progressive complexity):

1. **Phase 1 - Parametric Gaussian** (simplest, start here):

$$k_{\text{est}} = \text{Gaussian}(\sigma)$$

σ = learnable parameter initialized to 2.0

2. **Phase 2 - FFT-based** (if Phase 1 insufficient):

$$k_{\text{est}} = |\text{FFT}^{-1}(\text{FFT}(y) / \text{FFT}(x_{\text{pred}}))|$$

Apply threshold to prevent noise amplification

3. Phase 3 - Learned estimator (if needed):

$k_{est} = \text{Small CNN}(\text{concat}(y, x_{pred}))$

3-layer network, output 5×5 or 7×7 kernel

Why This Helps:

- Enforces physical consistency: sharp output re-blurred should match input gradients
- Complements existing edge loss (which only enforces sharpness, not blur physics)
- Implicit PSF estimation adapts to varying blur types
- Gradient domain naturally emphasizes edges while being robust to illumination changes

Implementation Details:

- Gradient computation: Sobel filters in x, y directions
- Gradient magnitude: $\sqrt{\nabla_x^2 + \nabla_y^2}$
- L2 loss for stability (vs. L1 which can cause unstable training)
- Only applied to decoder output (not intermediate $x \downarrow 8$)
- Weight: $\lambda_{blur_grad} = 10.0$ (similar scale to $\lambda_{edge} = 50$)

4.2.3 Phase-Enhanced Frequency Loss ($L_{freq_enhanced}$)

Motivation: DarkIR's FreMLP operates on Fourier amplitude to enhance illumination, but ignores phase. However, phase contains critical edge and structural information. Low-light images have degraded amplitude but relatively preserved phase - we should exploit both.

Formulation:

Decompose predictions and ground truth into frequency components:

$F_{pred} = \text{FFT}(I_{pred}), \quad F_{gt} = \text{FFT}(I_{gt})$

$A_{pred} = |F_{pred}|, \quad A_{gt} = |F_{gt}| \quad \# \text{ Amplitude}$

$\phi_{pred} = \angle F_{pred}, \quad \phi_{gt} = \angle F_{gt} \quad \# \text{ Phase}$

The enhanced loss combines both:

$L_{freq_enhanced} = \alpha \cdot ||A_{pred} - A_{gt}||_1 + \beta \cdot L_{phase}$

Phase Loss Options (to handle 2π wrapping):

1. **Cosine-Sine Representation** (numerically stable):

$$L_{\text{phase}} = ||\cos(\varphi_{\text{pred}}) - \cos(\varphi_{\text{gt}})||_1 + ||\sin(\varphi_{\text{pred}}) - \sin(\varphi_{\text{gt}})||_1$$

2. **Complex Distance** (mathematically elegant):

$$L_{\text{phase}} = ||\exp(i\varphi_{\text{pred}}) - \exp(i\varphi_{\text{gt}})||_1$$

3. **Phase Gradient** (more robust):

$$L_{\text{phase}} = ||\nabla\varphi_{\text{pred}} - \nabla\varphi_{\text{gt}}||_1$$

where ∇ = spatial gradient (less sensitive to wrapping than direct phase difference)

We will experimentally compare all three, likely favoring option (3) for stability.

Frequency Band Weighting: Phase is unreliable at very low frequencies (DC, near-DC) due to noise. We apply band-limited phase loss:

$$\text{mask} = (\text{freq}_x^2 + \text{freq}_y^2 > \text{threshold}) \quad \# \text{ High-pass filter}$$

$$L_{\text{phase}} = ||\text{mask} \cdot (\varphi_{\text{pred}} - \varphi_{\text{gt}})||_1$$

Why This Helps:

- Amplitude enhancement (existing DarkIR) + phase preservation (new) = complete frequency-domain supervision
- Phase encodes edges and textures that are critical for perceptual quality
- Prevents detail loss during illumination adjustment
- Multi-scale application naturally captures coarse (illumination) and fine (detail) structures

Implementation Details:

- Use torch.fft.rfft2 for efficiency (real FFT)
- Weights: $\alpha = 1.0$, $\beta = 0.1$ (phase is sensitive, use low weight initially)
- Apply at multiple scales: 256×256 and 512×512
- Only high-frequency phase (above 1/32 of Nyquist)
- Can be applied to both encoder (FreMLP features) and decoder output

4.3 Complete Training Objective

The final loss integrates physics-based components with DarkIR's existing losses:

$$L_{\text{total}} = \lambda_{\text{pixel}} \cdot L1(I_{\text{pred}}, I_{\text{gt}}) \\ + \lambda_{\text{percep}} \cdot \text{LPIPS}(I_{\text{pred}}, I_{\text{gt}})$$

$$\begin{aligned}
& + \lambda_{\text{edge}} \cdot ||\nabla I_{\text{pred}} - \nabla I_{\text{gt}}||_2^2 \\
& + \lambda_{\text{lol}} \cdot L1(\hat{x} \downarrow 8, x \downarrow 8) \\
& + \lambda_{\text{retinex}} \cdot L_{\text{retinex}}(I_{\text{pred}})[\text{NEW}] \\
& + \lambda_{\text{blur_grad}} \cdot L_{\text{blur_grad}}(I_{\text{pred}}, y_{\text{input}})[\text{NEW}] \\
& + \lambda_{\text{freq}} \cdot L_{\text{freq_enhanced}}(I_{\text{pred}}, I_{\text{gt}})[\text{ENHANCED}]
\end{aligned}$$

Loss Weighting Strategy:

Initial values (based on scale analysis):

- $\lambda_{\text{pixel}} = 1.0$, $\lambda_{\text{percep}} = 0.01$, $\lambda_{\text{edge}} = 50$, $\lambda_{\text{lol}} = 1.0$ (DarkIR defaults)
- $\lambda_{\text{retinex}} = 0.1$ (start conservative)
- $\lambda_{\text{blur_grad}} = 10.0$ (comparable to edge loss)
- $\lambda_{\text{freq}} = 0.05$ (phase is sensitive)

Progressive Integration (de-risk training instabilities):

1. Epochs 1-50: Baseline DarkIR losses only
2. Epochs 51-150: + L_{retinex} (validate stability)
3. Epochs 151-250: + $L_{\text{blur_grad}}$ (add deblurring physics)
4. Epochs 251-300: + $L_{\text{freq_enhanced}}$ (full method)

This staged approach allows us to:

- Isolate each loss component's contribution
- Detect training instabilities early
- Tune hyperparameters incrementally

Computational Overhead:

- L_{retinex} : ~2% overhead (max pooling + gradient computation)
- $L_{\text{blur_grad}}$: ~5% overhead (convolution + gradient)
- $L_{\text{freq_enhanced}}$: ~3% overhead (FFT already used in architecture)
- **Total**: <10% training time increase, negligible inference overhead

4.4 Experimental Design

4.4.1 Training Configuration

Datasets:

- Primary: LOLBlur (10,200 train, 1,800 test) with real sensor noise variant
- Validation: Hold out 10% of LOLBlur train for hyperparameter tuning

Training Details:

- Base architecture: DarkIR-m (3.31M parameters)
- Optimizer: AdamW ($\beta_1=0.9$, $\beta_2=0.9$, weight decay= $1e-3$)
- Learning rate: $5e-4 \rightarrow 1e-6$ (cosine annealing)
- Batch size: 32 (crop size 384×384)
- Total epochs: 300
- Estimated training time: 48-60 hours

Data Augmentation (following DarkIR):

- Random horizontal/vertical flips
- Random 90° rotations
- No additional augmentation (keep consistent with baseline)

4.4.2 Evaluation Protocol**Test Sets:**

1. LOLBlur test (1,800 pairs) - primary benchmark
2. Real-LOLBlur (482 images) - real-world generalization, no GT
3. LOLv2-Real (100 test pairs) - cross-dataset generalization
4. LOLv2-Synthetic (100 test pairs) - synthetic generalization
5. LSRW-Huawei, LSRW-Nikon - camera-specific robustness

Metrics:*With Ground Truth:*

- PSNR, SSIM (distortion metrics)
- LPIPS (perceptual quality)
- Edge preservation: Canny edge detection + IoU with GT edges

No Ground Truth (Real-LOLBlur):

- MUSIQ, NRQM, NIQE (no-reference quality)
- Naturalness score (via pre-trained quality estimator)

Efficiency:

- Parameters (M), MACs (G) - should match DarkIR baseline
- Training time per epoch
- Inference time per image (at least 256×256, 512×512)

4.4.3 Possible Ablation Studies

Primary Ablations (8 training runs):

Experiment	L_retinex	L_blur_grad	L_freq_enh	Purpose
0. Baseline	✗	✗	✗ (amplitude only)	DarkIR reproduction
1. +Retinex	✓	✗	✗	Illumination physics
2. +Blur	✗	✓	✗	Deblurring physics
3. +Freq	✗	✗	✓	Phase preservation
4. Retinex+Blur	✓	✓	✗	Combined spatial
5. Retinex+Freq	✓	✗	✓	Illumination+detail
6. Blur+Freq	✗	✓	✓	Deblur+detail
7. Full (ours)	✓	✓	✓	Complete method

Each run: 300 epochs, same random seed for first 50 epochs (fair comparison)

4.4.4 Baseline Comparisons

Methods:

- DarkIR-v1 (our baseline, reproduced)
- LEDNet (previous SOTA joint method)
- RetinexFormer (LLIE specialist)
- NAFNet, Restormer (retrained on LOLBlur)

Comparison Metrics:

- Quantitative: PSNR, SSIM, LPIPS on all testsets
- Qualitative: Side-by-side visual comparisons (10-15 challenging images)
- Efficiency: Params, MACs, inference time
- Generalization: Performance gap between LOLBlur and LOLv2/LSRW

4.4.6 Success Criteria

Minimum Viable:

- +0.5 dB PSNR improvement on LOLBlur test
- At least two of three losses show positive contribution in ablations
- No training instabilities or divergence
- Qualitative improvements visible in side-by-side comparisons
- <10% training time overhead

Target:

- +0.8-1.2 dB PSNR on LOLBlur
- -0.02 to -0.05 LPIPS improvement
- Better generalization: +0.5-1.0 dB on LOLv2, LSRW (zero-shot)
- All three losses contribute positively (validated via ablations)
- Clear interpretability gains (visualizations show physically meaningful patterns)
- Demonstrated transferability: Apply losses to RetinexFormer or NAFNet, show gains

5. Conclusion

This research proposal presents a focused investigation into physics-guided loss functions for low-light image restoration. By integrating Retinex decomposition principles, blur-aware gradient matching, and phase-enhanced frequency constraints into the training objectives, we aim to improve both the quality and physical plausibility of restored images.

The proposed approach represents a principled bridge between classical image formation models and modern deep learning, offering several advantages over purely data-driven methods: improved generalization through physics-based regularization, better interpretability via explicit decompositions, and enhanced robustness to out-of-domain degradations.

The physics-based loss functions, if developed successfully, could provide a strong foundation for the subsequent research, which will integrate architectural innovations (multi-scale processing, adaptive attention) with these loss functions to create a comprehensive restoration framework.